SPEECHMASTER: CONFIDENCE-ROUTED SELF-SUPERVISED ASR WITH DISCRETE UNIT BUDGETING

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ABSTRACT

Self-supervised speech models are highly effective for low-resource ASR, yet using the strongest model on every utterance wastes computation and hides the representation/cost tradeoff. We present SpeechMaster, a budget-aware ASR framework that composes heterogeneous SSL representations instead of selecting a single encoder. SpeechMaster first decodes with a fast Wav2Vec2 CTC branch, computes an utterance-level CTC uncertainty score from non-blank posterior peakiness and entropy, and routes only uncertain utterances to a stronger HuBERT branch. A separate HuBERT discrete-unit auditor measures token rate and bitrate, linking recognition accuracy to representation compactness. On LibriSpeech clean validation slices, routing only 25% of utterances reduces WER from 2.47% to 1.88%, while the full HuBERT branch reaches 1.58%. An oracle route obtains 1.19% WER, confirming branch complementarity.

***Index Terms*—** ASR, self-supervised speech, uncertainty routing, HuBERT, Wav2Vec2

1. INTRODUCTION

Low-resource ASR is limited by the amount of transcribed speech, while modern self-supervised learning (SSL) encoders reduce this dependence by pretraining on unlabeled audio. Wav2Vec2, HuBERT, and WavLM learn complementary speech representations: contrastive contextual features, masked hidden-unit prediction, and denoising-enhanced masked prediction. A common project design is to choose one SSL encoder and report WER. This leaves two practical questions unanswered: when is a lightweight SSL recognizer already sufficient, and when should a stronger recognizer be invoked?

We propose SpeechMaster, a composed ASR framework for this setting. Its key idea is to treat SSL recognizers as a routed ensemble under a compute budget. The fast branch gives a first transcription and a CTC confidence estimate. The router sends only low-confidence utterances to a stronger SSL branch. In parallel, a discrete-unit auditor clusters HuBERT hidden states and reports bitrate, so the same system studies accuracy, speed, and representation compression. The contributions are: (1) a confidence-routed SSL-ASR framework, (2) a CTC uncertainty score requiring no extra labels or training, (3) a discrete-unit budget auditor, and (4) reproducible LibriSpeech experiments with WER, CER, RTF, token rate, and bitrate.

2. SPEECHMASTER METHOD

2.1. Two-Branch SSL Recognizer

Given waveform x, a pretrained SSL encoder f_θ produces hidden states $H = f_\theta(x)$. A CTC projection maps frames to vocabulary logits:

$$p_\theta(y_t \mid x) = \mathrm{softmax}(WH_t + b). \tag{1}$$

SpeechMaster uses a fast Wav2Vec2 branch F and a stronger HuBERT branch S. The fast branch is always evaluated. The strong branch is evaluated only if the router decides that the fast branch is uncertain.

2.2. CTC Uncertainty Router

For each non-blank frame of the fast branch, SpeechMaster measures posterior peakiness and entropy. Let q_t be the CTC posterior and $\mathcal{T}$ be the frames whose argmax is not blank. The confidence score is

$$c(x) = \frac{1}{|\mathcal{T}|}\sum_{t\in\mathcal{T}} \max_k q_t(k) - \lambda \frac{1}{|\mathcal{T}|}\sum_{t\in\mathcal{T}} H(q_t), \tag{2}$$

with $\lambda = 0.03$. For a routing budget B, utterances with the lowest $c(x)$ are sent to HuBERT. This gives a tunable continuum between fast-only decoding and strong-only decoding:

$$\hat{y}(x) = \begin{cases} S(x), & x \in \mathrm{BottomB}(c), \\ F(x), & \text{otherwise.} \end{cases} \tag{3}$$

2.3. Layer and Unit Auditors

SpeechMaster includes two auditors for analysis. The layer auditor decodes Wav2Vec2 final states and a last-four-layer average through the same CTC head, testing whether the projection is calibrated to a particular depth. The unit auditor clusters HuBERT hidden states with MiniBatchKMeans and computes token rate and bitrate:

Table 1. SpeechMaster budgeted routing results.

Route	Routed utt.	WER	CER	RTF
0%	0	2.474	0.945	0.002
25%	32	1.877	0.633	0.003
50%	64	1.792	0.603	0.003
75%	96	1.706	0.553	0.003
100%	128	1.578	0.493	0.004
Oracle	-	1.195	0.382	-

$$\text{bitrate} = r_{\text{tok}} \log_2 K. \quad (4)$$

Consecutive duplicate units are collapsed to estimate a compact content stream.

3. EXPERIMENTAL SETUP

3.1. Data and Models

Experiments use LibriSpeech clean validation slices. The repository also provides a low-resource fine-tuning script for train-clean-100 subsets. The fast branch is Wav2Vec2 base fine-tuned on LibriSpeech; the strong branch is HuBERT large fine-tuned on LibriSpeech; the unit auditor uses HuBERT base. Audio is resampled to 16 kHz.

3.2. Metrics

ASR quality is measured by WER and CER after LibriSpeech-style text normalization. Inference efficiency is measured by real-time factor (RTF). Discrete units are evaluated by codebook size, token rate, deduplicated token rate, bitrate, and deduplicated bitrate. These metrics correspond to the assignment requirements and make the recognition/compression tradeoff explicit.

4. RESULTS

Table 1 is the main SpeechMaster result. Routing only the lowest-confidence 25% of utterances recovers most of the gain over Wav2Vec2, reducing WER by 24.1% relative while adding a small amount of cascade cost. The oracle row shows the complementarity ceiling if routing decisions were perfect.

Table 2 reports ASR performance. Lower WER/CER is better. The layer-fusion row tests whether averaging SSL layers improves the CTC decision surface without training a new decoder.

Table 3 summarizes discrete-unit compression. Increasing K raises the nominal bitrate, while deduplication reduces token rate by merging stable frame-level assignments.

Table 2. ASR results on LibriSpeech clean validation slices.

System	Rep.	N	WER	CER	RTF
wav2vec2-base-960h	final	128	2.474	0.945	0.003
wav2vec2-base-960h	last4	128	2.858	1.026	0.003
hubert-large-ls960-ft	final	128	1.578	0.493	0.005

Table 3. HuBERT discrete-unit statistics.

System	K	Tok/s	Dedup tok/s	Bit/s	Dedup bit/s
hubert-base-ls960	50	49.86	25.42	281.41	143.45
hubert-base-ls960	100	49.86	28.15	331.28	187.05
hubert-base-ls960	200	49.86	29.51	381.14	225.58
hubert-base-ls960	500	49.86	33.31	447.05	298.68

This makes it possible to compare a continuous CTC system against compact speech-token representations.

4.1. Ablation and Error Analysis

The ablations cover model family, layer fusion, routing budget, and codebook size. HuBERT has lower WER than Wav2Vec2, but the oracle route is better than either individual branch, indicating non-identical errors. The last-four-layer average is worse than the final Wav2Vec2 layer, suggesting the pretrained CTC head is calibrated to final-layer content. Increasing codebook size raises bitrate while reducing clustering distortion, so compact speech units should be chosen according to the downstream cost budget.

5. CONCLUSION

SpeechMaster combines fast and strong SSL recognizers through CTC uncertainty routing, then audits the representation cost with discrete HuBERT units. The results show that a small routed fraction gives a large WER reduction, while the oracle route reveals remaining complementarity. This supports the central claim: low-resource SSL-ASR should optimize accuracy, inference budget, and representation bitrate jointly instead of treating encoder selection as a single-model decision.

6. REFERENCES

7. REFERENCES

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